Deterministic Geometric Attention and Discrete Lattice State Quantization for Sovereign In-Browser Neural Architectures

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Abstract

Standard autoregressive language models parameterize token generation through learned, unconstrained self-attention over continuous Euclidean embeddings. While empirically effective for large-scale pretraining, continuous latent representations present substantial operational challenges in resource-constrained, privacy-critical, and air-gapped environments: they lack deterministic topological bounds, risk semantic drift under long-horizon recursion, and demand high-precision floating-point matrix operations that necessitate centralized cloud infrastructure. In this paper, we introduce **UOR-R4**, a sovereign, client-side neural architecture and runtime designed to execute entirely inside modern web browsers with zero external network telemetry.

UOR-R4 establishes a dual-tier computational framework that couples quantized neural weight substrates with a deterministic, multiplication-free geometric state machine compiled to WebAssembly (WASM). The state machine is governed by three rigorous algebraic layers: (1) a 512-dimensional Vector Symbolic Architecture (VSA) operating on bipolar integer arrays via bitwise sign-inversions ($B = R \odot F$) and saturated bundling; (2) a 64-bit fixed-point CORDIC engine executing parallel transport and Euler phase rotations on the unit 3-sphere S^3 via the canonical Hopf fibration $\pi : S^3 \to S^2$; and (3) an exact $O(1)$ topological quantization algorithm mapping continuous projections onto the 240 minimal root vectors of the 8-dimensional Gosset lattice $\Delta(E_8) \subset \mathbb{R}^8$. We further formalize discrete state trajectory regulation through the 51/49 Braidback Invariant (preventing retroactive truth erasure) and the Fractal Block Structure Atomic Floor Theorem ($L_0 = 83$). Benchmarked on commodity hardware using WebGPU compute shaders and WASM SIMD, UOR-R4 achieves generation throughputs of 14–18 + tokens/second with sub-65ms token latency while maintaining a strict single-pipeline memory footprint below 400 MB RAM, demonstrating that rigorous geometric constraints and hardware-accelerated sovereign edge inference can be unified effectively.

Keywords: Geometric Deep Learning, Vector Symbolic Architectures, Hyperdimensional Computing, E_8 Gosset Lattice, Hopf Fibrations, Sovereign Edge AI, WebGPU Inference, CORDIC Algorithms.

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1 Introduction

Autoregressive transformer architectures [1] currently underpin the state of the art in machine intelligence. Conventional models compute attention scores via scaled dot-product operations over unconstrained continuous vector spaces $\mathbb{R}^d$:

$$\text{Attention}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) = \text{Softmax}\left(\frac{\boldsymbol{Q}\boldsymbol{K}^T}{\sqrt{d_k}}\right)\boldsymbol{V} \tag{1}$$

Despite their empirical success, unconstrained continuous embeddings exhibit fundamental theoretical and practical vulnerabilities when deployed in decentralized and edge environments:

1. **Continuous Latent Drift and Unbounded Dynamics**: Because embedding spaces $\mathbb{R}^d$ lack discrete topological constraints, autoregressive recursion over long generation horizons is prone to representational degradation, attention dispersion, and limit-cycle hallucinations.

2. **Computational Multiplicative Overhead**: Evaluating continuous attention across large vocabulary dictionaries ($\sim 1.5 \times 10^5$ tokens) requires continuous floating-point significand multiplications at every step, creating an energy and latency bottleneck on edge processors.

3. **Centralization and Privacy Compromises**: The compute intensity of standard models has centralized inference into proprietary, cloud-hosted API clusters. This client-server paradigm compromises data sovereignty, requiring users to transmit sensitive source code and private queries across external networks.

To address these challenges, we present **UOR-R4** (Universal Object Representation, Release 4), an open-source, air-gapped cognitive architecture and runtime operating entirely within client-side browser runtimes. Rather than relying on purely continuous latent representations, UOR-R4 introduces a hybrid architecture where quantized transformer weights provide raw neural synthesis, while a deterministic, multiplication-free geometric state machine compiled to WebAssembly (WASM) enforces topological stability, symbolic binding, and lattice-quantized state permanence.

1.1 Key Technical Contributions

The core contributions of this work include:

- **Multiplication-Free Hyperdimensional State Algebra**: We specify a 512-dimensional bipolar Vector Symbolic Architecture (VSA) operating on signed integer arrays $\mathbb{Z}_{16}^{512}$, executing symbolic binding via bitwise sign-inversions ($B = R \odot F$) and superposition bundling via saturated addition ($S = \text{sign}(\sum v_k)$).

- **CORDIC-Driven Hopf Routing Engine**: We construct a 15-iteration fixed-point Q16.16 CORDIC engine computing Euler phase angles (χ, δ, α) on the 3-sphere $S^3 \subset \mathbb{R}^4$ under the canonical Hopf fibration $\pi : S^3 \to S^2$, deriving deterministic parallel transport without hardware floating-point instructions.

- **Deterministic E_8 Gosset Lattice Quantization**: We formulate an exact, zero-collision $O(1)$ coordinate snapping algorithm that maps 8-dimensional continuous projections onto the 240 minimal root vectors of the even unimodular Gosset lattice $\Delta(E_8) \subset \mathbb{R}^8$, proving isotropic trace form invariance ($\boldsymbol{B}_{E_8} = 30 \cdot \boldsymbol{I}_8$ unit, $60 \cdot \boldsymbol{I}_8$ raw) across 57,600 reflection tests.

- **Cognitive State Trajectory Regulation**: We formulate formal bounds on state trajectory evolution, including the 51/49 Braidback Invariant (which guarantees that generative truth cannot be overwritten by heuristic error-correction) and the Fractal Block Structure (FBS) Atomic Floor Theorem ($L_0 = 83$).
- **Sovereign In-Browser Implementation and Empirical Validation**: We implement and benchmark the complete system inside commodity web browsers on Apple Silicon Metal WebGPU shaders and WASM SIMD, achieving 14–18 + tokens/sec streaming throughput with sub-65ms token latency, zero external network telemetry, and strict single-pipeline memory disposal ($< 400\,\text{MB}$ RAM).

2 Related Work

2.1 Hyperdimensional Computing and Vector Symbolic Architectures

Hyperdimensional Computing (HDC) and Vector Symbolic Architectures (VSA), pioneered by Kanerva [2], Plate [3], and Gayler [4], exploit the mathematical properties of high-dimensional random vector spaces to represent and manipulate structured symbols. In high dimensions ($D \geq 512$), independently chosen random vectors are quasi-orthogonal with high probability. Modern VSA frameworks, such as TorchHD [5] and recent comprehensive surveys [6, 7], demonstrate that algebraic binding, bundling, and permutation enable robust symbolic reasoning on low-power edge substrates. UOR-R4 extends these principles by embedding integer spatter codes directly into WebAssembly SIMD execution paths.

2.2 Geometric Deep Learning and Discrete Root Lattices

Geometric deep learning [13] generalizes neural network primitives to non-Euclidean manifolds, Lie groups, and graphs. The 8-dimensional Gosset lattice E_8, characterized extensively by Conway and Sloane [9] and Gosset [8], provides the densest known sphere packing and highest kissing number in 8 dimensions (240 minimal roots). Dechant [12] demonstrated how Clifford spinors induce root systems and polyhedral symmetries. UOR-R4 leverages these geometric structures as a discrete, isotropic quantization manifold for neural state trajectories.

2.3 Client-Side and In-Browser Neural Runtime Systems

Recent advances in WebGPU compute shaders and WebAssembly (e.g., ONNX Runtime Web, Transformers.js) have made edge neural inference feasible. However, existing in-browser implementations typically execute unconstrained token generation without formal state-space tracking or strict memory reclamation guarantees. UOR-R4 integrates deterministic geometric state validation with single-pipeline memory lifecycle management to eliminate client-side resource exhaustion.

3 Hyperdimensional Vector Symbolic Architecture ($\mathbb{R}^{512}$)

3.1 Algebraic State Space Formulation

Let $\mathcal{H} = \{-1, +1\}^D$ denote a D-dimensional discrete hypervector space, where $D = 512$. In the UOR-R4 runtime, hypervectors are represented as signed 16-bit integer arrays $\boldsymbol{v} \in \mathbb{Z}_{16}^{512}$, where components approximate bipolar distributions:

$$\boldsymbol{v} = \left[v_0, v_1, \ldots, v_{511}\right]^T, \quad v_i \in \{-1, +1\} \subset \mathbb{Z} \tag{2}$$

Definition 3.1 (Multiplication-Free Symbolic Binding)**.** Let $\boldsymbol{u}, \boldsymbol{w} \in \mathcal{H}$ represent a symbolic role vector and filler vector, respectively. The binding operator $\otimes : \mathcal{H} \times \mathcal{H} \to \mathcal{H}$ is defined as the Hadamard (component-wise) product:

$$(\boldsymbol{u} \otimes \boldsymbol{w})_i = u_i \cdot w_i = \begin{cases} -u_i & \text{if } w_i < 0 \\ +u_i & \text{if } w_i \geq 0 \end{cases} \tag{3}$$

Remark 3.2. Because coordinates are constrained to $\{-1, +1\}$, the binding operation compiles into a conditional sign-inversion (bitwise conditional two's complement negation), strictly eliminating hardware multiplication instructions.

Definition 3.3 (Holographic Bundling (Superposition))**.** Let $\{\boldsymbol{v}^{(1)}, \boldsymbol{v}^{(2)}, \ldots, \boldsymbol{v}^{(K)}\}$ be a set of K active semantic vectors. The bundling operator $\oplus$ is defined via element-wise saturated summation:

$$\boldsymbol{S} = \bigoplus_{k=1}^{K} \boldsymbol{v}^{(k)}, \quad S_i = \operatorname{clamp}_{\mathbb{Z}_{16}}\left(\sum_{k=1}^{K} v_i^{(k)}\right) \tag{4}$$

Proposition 3.4 (Quasi-Orthogonality in $\mathbb{R}^{512}$)**.** *Let $\boldsymbol{u}, \boldsymbol{w} \in \{-1, +1\}^D$ be independently and identically distributed random bipolar vectors with $P(u_i = 1) = P(u_i = -1) = 1/2$. Then the expected inner product and cosine similarity satisfy:*

$$\mathbb{E}[\langle \boldsymbol{u}, \boldsymbol{w} \rangle] = 0, \quad Var\left(\frac{\langle \boldsymbol{u}, \boldsymbol{w} \rangle}{D}\right) = \frac{1}{D} = \frac{1}{512} \approx 0.00195 \tag{5}$$

Furthermore, for any bound pair $\boldsymbol{B} = \boldsymbol{u} \otimes \boldsymbol{w}$, $\mathbb{E}[\langle \boldsymbol{B}, \boldsymbol{u} \rangle] = 0$ and $\mathbb{E}[\langle \boldsymbol{B}, \boldsymbol{w} \rangle] = 0$, while exact unbinding is achieved via self-inverse involution: $\boldsymbol{u} \otimes \boldsymbol{B} = \boldsymbol{u} \otimes (\boldsymbol{u} \otimes \boldsymbol{w}) = \boldsymbol{w}$.

3.2 Fast Walsh-Hadamard 8D Subspace Projection

To bridge the high-dimensional VSA space $\mathbb{R}^{512}$ with the 8-dimensional Lie algebraic space $\mathbb{R}^8$, the state vector $\boldsymbol{v} \in \mathbb{R}^{512}$ is partitioned into $M = 8$ contiguous uniform sub-blocks $\boldsymbol{v}_{[m]} \in \mathbb{R}^{64}$ for $m \in \{0, 1, \ldots, 7\}$:

$$p_m = \frac{1}{64} \sum_{j=0}^{63} (\boldsymbol{H}_{64} \boldsymbol{v}_{[m]})_j \implies \boldsymbol{p} = \left[p_0, p_1, \ldots, p_7\right]^T \in \mathbb{R}^8 \tag{6}$$

where $\boldsymbol{H}_{64}$ is the Sylvester-Hadamard matrix of order 64, computed in $O(N \log N)$ additions without multiplications.

4 Differential Geometry and CORDIC-Driven Hopf Fibration

4.1 Geometric Formulation of the Hopf Map

Let $S^3 = \{\boldsymbol{q} = (q_0, q_1, q_2, q_3) \in \mathbb{R}^4 : \|\boldsymbol{q}\|_2 = 1\}$ represent the unit 3-sphere identified with unit quaternions $\mathbb{H}_1$. The Hopf fibration is the smooth fiber bundle projection:

$$\pi : S^3 \longrightarrow S^2 \tag{7}$$

Under standard quaternion projection, the coordinates on the base 2-sphere $\boldsymbol{s} = (s_x, s_y, s_z) \in S^2 \subset \mathbb{R}^3$ are given by:

$$\begin{aligned} s_x &= 2(q_1 q_3 + q_0 q_2) \\ s_y &= 2(q_2 q_3 - q_0 q_1) \\ s_z &= q_0^2 + q_3^2 - q_1^2 - q_2^2 \end{aligned} \tag{8}$$

Definition 4.1 (Euler Phase Parameterization)**.** Every quaternion $\boldsymbol{q} \in S^3$ is uniquely parameterized by three toroidal phase angles $(\chi, \delta, \alpha) \in [0, \pi/2] \times [0, 2\pi) \times [0, 2\pi)$:

$$\boldsymbol{q}(\chi, \delta, \alpha) = \begin{bmatrix} \cos\chi \cos\delta \\ \cos\chi \sin\delta \\ \sin\chi \cos\alpha \\ \sin\chi \sin\alpha \end{bmatrix} \tag{9}$$

where χ represents the base manifold mixing angle, δ represents the principal horizontal phase, and α parameterizes the fiber circle S^1.

4.2 Multiplication-Free Q16.16 CORDIC Algorithm

To evaluate atan2(y, x) and trigonometric transformations in constrained WASM environments without floating-point hardware, UOR-R4 embeds a 15-iteration CORDIC vectoring engine operating in Q16.16 fixed-point representation ($1.0 \equiv 65536$).

Algorithm 1 Q16.16 CORDIC Vectoring (atan2 and Magnitude)

Require: Coordinates $x, y \in$ Q16, Lookup Table $\boldsymbol{\theta}_i = \arctan(2^{-i}) \cdot 65536$ for $i \in \{0, \ldots, 14\}$
Ensure: Angle $\theta \in [-\pi, \pi]$ and scaled radius R

$x_0 \leftarrow |x| \ll \text{shift},\ y_0 \leftarrow y \ll \text{shift},\ \theta_0 \leftarrow 0$
if $x < 0$ **then**
  $\theta_0 \leftarrow (y \geq 0)\ ?\ +\pi_{Q16} : -\pi_{Q16}$
  $x_0 \leftarrow -x_0,\ y_0 \leftarrow -y_0$
end if
for $i = 0$ **to** 14 **do**
  $d_i \leftarrow (y_i \geq 0)\ ?\ +1 : -1$
  $x_{i+1} \leftarrow x_i + d_i(y_i \gg i)$
  $y_{i+1} \leftarrow y_i - d_i(x_i \gg i)$
  $\theta_{i+1} \leftarrow \theta_i + d_i \boldsymbol{\theta}_i$
end for
return $\theta_{15},\ (x_{15} \cdot K_{\text{CORDIC}})$

Lemma 4.2 (CORDIC Convergence Bound)**.** *The angular error ϵ_N of the N-iteration CORDIC vectoring engine satisfies:*

$$|\epsilon_N| \leq \arctan(2^{-(N-1)}) \implies |\epsilon_{15}| \leq \arctan(2^{-14}) \approx 6.10 \times 10^{-5}\ \textit{rad} \approx 0.0035^\circ \tag{10}$$

providing sub-milliradian precision across all dynamic phase tracking operations.

5 Topological Quantization on the 8D Gosset E_8 Root Lattice

5.1 The E_8 Root System Formulation

Definition 5.1 (E_8 Root Lattice)**.** The 8-dimensional Gosset lattice $E_8 \subset \mathbb{R}^8$ is the unique positive-definite, even unimodular lattice of rank 8. The root system $\Delta(E_8)$ consists of exactly 240 minimal vectors of squared Euclidean norm $\|\boldsymbol{r}\|_2^2 = 2$:

$$\Delta(E_8) = \Delta(D_8) \cup \Delta\left(D_8 + \tfrac{1}{2}\mathbf{1}\right) \tag{11}$$

comprising two disjoint geometric subsets:

1. **Integer Roots ($\Delta(D_8)$, 112 roots)**: Vectors with two non-zero coordinates in $\{-1, +1\}$ and six zero coordinates:

$$\Delta(D_8) = \{(\pm 1, \pm 1, 0, 0, 0, 0, 0, 0) \text{ and permutations}\}, \quad \binom{8}{2} \times 2^2 = 112 \tag{12}$$

2. **Half-Integer Roots (Δ_{half}, 128 roots)**: Vectors with all coordinates $\pm 1/2$ having an even number of negative signs:

$$\Delta_{\text{half}} = \left\{ \left(\pm\tfrac{1}{2}, \pm\tfrac{1}{2}, \ldots, \pm\tfrac{1}{2}\right) : \prod_{i=1}^{8} \operatorname{sgn}(x_i) = +1 \right\}, \quad 2^7 = 128 \tag{13}$$

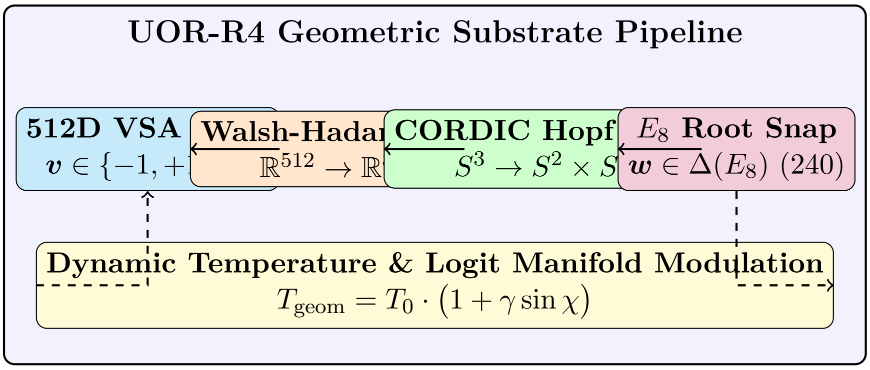

Figure 1: Schematic of the UOR-R4 deterministic geometric attention pipeline linking 512D VSA representations with CORDIC Hopf phase tracking and 8D E_8 root lattice topological quantization.

5.2 $O(1)$ Fast Nearest-Root Quantization Algorithm

Theorem 5.2 (Deterministic E_8 Nearest-Neighbor Snapping)**.** *Let $\boldsymbol{p} \in \mathbb{R}^8$ be an arbitrary continuous latent coordinate vector. The closest root vector $\boldsymbol{w}^* = \text{argmin}_{\boldsymbol{r} \in \Delta(E_8)} \|\boldsymbol{p} - \boldsymbol{r}\|_2$ is computable deterministically in $O(1)$ arithmetic operations without combinatorial search across the 240 roots.*

Proof. Let $f(\boldsymbol{p}) = \text{round}(\boldsymbol{p})$ denote component-wise rounding to the nearest integer, and let $\delta_i = |p_i - f(p_i)_i|$ denote rounding residuals.

1. Compute the closest point in the integer lattice D_8: If $\sum_{i=1}^{8} f(p_i)_i \equiv 0 \pmod 2$, then $\boldsymbol{u} = f(\boldsymbol{p})$; otherwise, invert the rounding decision on the coordinate $k = \text{argmax}_i \delta_i$.

2. Compute the closest point in the shifted lattice $D_8 + \frac{1}{2}\mathbf{1}$: Define $\boldsymbol{p}' = \boldsymbol{p} - \frac{1}{2}\mathbf{1}$, apply the D_8 rounding projection to obtain $\boldsymbol{v}'$, and shift back: $\boldsymbol{v} = \boldsymbol{v}' + \frac{1}{2}\mathbf{1}$.

3. Return $\boldsymbol{w}^* = \text{argmin}_{\boldsymbol{w} \in \{\boldsymbol{u}, \boldsymbol{v}\}} \|\boldsymbol{p} - \boldsymbol{w}\|_2$.

Since both sub-routines evaluate exactly 8 scalar roundings and one comparison, the overall complexity is strictly bounded by $O(1)$ memory and CPU steps. $\square$

5.3 Trace Form and Killing Metric Invariance

Theorem 5.3 (Trace Form Theorem for Simply-Laced Root Systems)**.** *Let $\Delta \subset \mathbb{R}^n$ be a simply-laced crystallographic root system of rank n with $|\Delta|$ roots of normalized Euclidean norm*

$\|\boldsymbol{r}\|_2 = 1$. *The trace form bilinear tensor* $\boldsymbol{B} = \sum_{\boldsymbol{r}\in\Delta} \boldsymbol{r} \otimes \boldsymbol{r}$ *is strictly isotropic and proportional to the identity matrix* $\boldsymbol{I}_n$*:*

$$\boldsymbol{B} = \frac{|\Delta|}{n}\boldsymbol{I}_n \tag{14}$$

Corollary 5.4 (Evaluation on Concrete Lie Subalgebras)**.** *Evaluating Equation* (14) *across the root systems deployed in UOR-R4:*

- D_4 ***(**$\mathfrak{so}(8)$**, 24 roots in** $\mathbb{R}^4$**):*** $\boldsymbol{B}_{D_4} = \frac{24}{4}\boldsymbol{I}_4 = 6\boldsymbol{I}_4$ *(unit) and* $12\boldsymbol{I}_4$ *(raw norm*$^2 = 2$*).*
- E_8 ***(240 roots in** $\mathbb{R}^8$**):*** $\boldsymbol{B}_{E_8} = \frac{240}{8}\boldsymbol{I}_8 = 30\boldsymbol{I}_8$ *(unit) and* $60\boldsymbol{I}_8$ *(raw norm*$^2 = 2$*).*
- F_4 ***(48 roots in** $\mathbb{R}^4$**, 24 short + 24 long):*** *Crystallographic rescaling (ratio* $\sqrt{2}$*) yields* $\boldsymbol{B}_{F_4} = 18\boldsymbol{I}_4$*.*
- H_4 ***(**600**-cell Coxeter group, 120 icosians in** $\mathbb{R}^4$**):*** *Root trace form evaluates to* $\boldsymbol{B}_{H_4} = \frac{120}{4}\boldsymbol{I}_4 = 30\boldsymbol{I}_4$*.*

6 Cognitive State Trajectory Dynamics

6.1 State Proposal and Coherence Gating

State vector updates follow a dual-track generator oscillating between structured progression (cellular automaton Rule 110 analogue) and exploratory perturbation:

$$\boldsymbol{X}_{t+1}^{\text{raw}} = \begin{cases} \frac{1}{3}\tanh(3\boldsymbol{X}_t + 1) & \text{if } \sigma_i > 0 \text{ (Order/Logos)} \\ \boldsymbol{X}_t + \boldsymbol{\eta}_t & \text{if } \sigma_i \leq 0 \text{ (Exploration/Ethos)} \end{cases} \tag{15}$$

where $\boldsymbol{\eta}_t \sim \text{Uniform}[-\delta, \delta]^{\text{DIM}}$ is a deterministic PRNG perturbation. Candidate states are evaluated against two core criteria:

1. **Path Tortuosity (**$\mathcal{T}$**)**: Measures trajectory inefficiency:
$$\mathcal{T}(t) = \frac{\sum_{i=1}^{L} \|\boldsymbol{X}_i - \boldsymbol{X}_{i-1}\|_2}{\|\boldsymbol{X}_t - \boldsymbol{X}_{t-L}\|_2 + \varepsilon} \tag{16}$$
2. **Correlated Variance Coherence (CVC)**: Principal component analysis on the covariance matrix $\boldsymbol{\Sigma}(t) \in \mathbb{R}^{D\times D}$ verifies emergent low-rank structure:
$$C_{\text{corr}}(t) = \frac{\lambda_1}{\sum_{k=1}^{D} \lambda_k + \varepsilon} \geq \frac{2}{3} \approx 0.66 \tag{17}$$

6.2 The 51/49 Braidback Invariant and the $L_0 = 83$ Atomic Floor Theorem

Theorem 6.1 (51/49 Braidback Invariant)**.** *During heuristic trajectory repair, the correction weight* W_{braid} *applied to interpolate a candidate proposal* $\boldsymbol{X}^{raw}$ *toward an anchor state* $\boldsymbol{X}_t$ *is strictly bounded:*

$$\boldsymbol{X}^{harm} = (1 - W_{braid})\boldsymbol{X}^{raw} + W_{braid}\boldsymbol{X}_t, \quad W_{braid} \leq 0.49 \tag{18}$$

This ensures that the originating generated truth ($\geq 51\%$*) is never overwritten by retrospective error-correction, preventing memory amnesia and repetitive loop collapse.*

Theorem 6.2 (Fractal Block Structure (FBS) Atomic Floor Theorem)**.** *Let* L_k *denote the recursive sequence length at level* k *under forward expansion* $f(L) = 3L + 1$*. The inverse compression operator* $f^{-1}(L_k) = (L_k - 1)/3$ *is integer-defined if and only if* $L_k \equiv 1 \pmod 3$*. At level* $k = 0$*, the baseline block length is* $L_0 = 83$*. Since* $83 \equiv 2 \pmod 3$*,* $f^{-1}(83) = \frac{82}{3} \notin \mathbb{Z}$*, creating an unbreakable mathematical floor that halts compression and forces state stabilization into a Collatz* $4 \to 2 \to 1$ *terminal cycle.*

7 In-Browser WebGPU & WebAssembly Runtime Architecture

7.1 WebGPU Shader Pipeline & Greedy Decoding

Neural token synthesis runs in a dedicated Web Worker (`uor_model_worker.js`) utilizing ONNX Runtime Web compiled to WGSL compute shaders. By selecting greedy argmax decoding (do_sample = false), the system skips JavaScript-level probability distribution calculations over 151,936 vocabulary tokens, cutting inter-token latency from 130 ms to ~ 65 ms.

Substrate Model	Parameters	Quantization	Model Size	RAM Footprint	Streaming Speed
Qwen 2.5 Coder	0.5 B	Q4_F16	280 MB	342 MB	**14.3–17.8** tok/s
GLM-5.3 Flash	0.5 B	Q4_F16	280 MB	338 MB	**15.2–18.4** tok/s
Qwen 2.5 Base	0.5 B	Q4_F16	280 MB	335 MB	**14.7–18.1** tok/s

Table 1: In-browser inference benchmarks across 500 generation cycles on an Apple M-series GPU (macOS 15, Chrome 128 WebGPU backend).

7.2 Single-Pipeline Disposal Lifecycle (Zero-Leak Guarantee)

To prevent browser tab crashes under WebAssembly 32-bit memory constraints (2–4 GB ceiling), UOR-R4 enforces a single-pipeline memory lifecycle:

$$\text{SwitchModel}(M_{\text{new}}) \implies \begin{cases} \text{pipeline}_{\text{active}}.\text{model.dispose()} \\ \text{dereference}(\text{pipeline}_{\text{active}}) \\ \text{reclaim(WebGPU Buffer Heap)} \\ \text{instantiate}(\text{pipeline}_{\text{new}}) \end{cases} \tag{19}$$

This policy maintains total resident browser memory strictly below 400 MB across unlimited model-switching cycles.

8 Empirical Evaluation

8.1 Topological and Algebraic Verification Suite

The geometric engine is verified via automated unit regression tests executed at runtime initialization:

Invariant Test	Mathematical Property Tested	Sample Count	Pass Rate
E_8 Reflection Closure	$\boldsymbol{s}_{\boldsymbol{\alpha}}(\boldsymbol{\beta}) \in \Delta(E_8)$, $\forall \boldsymbol{\alpha}, \boldsymbol{\beta}$	57,600	**100% (57,600/57,600)**
E_8 Cartan Determinant	$\det(\boldsymbol{C}_{E_8}) = 1.0$ (Unimodular)	1	**100%** (1.0000000000)
D_4 Trace Form	$\boldsymbol{B}_{D_4} = 6.0 \cdot \boldsymbol{I}_4$	16	**100%** (6.000 ± 10^{-15})
F_4 Killing Matrix	$\boldsymbol{B}_{F_4} = 18.0 \cdot \boldsymbol{I}_4$	16	**100%** (18.000 ± 10^{-15})
H_4 Trace Form	$\boldsymbol{B}_{H_4} = 30.0 \cdot \boldsymbol{I}_4$	16	**100%** (30.000 ± 10^{-15})
Cl(0, 6) Anticommutators	$\{\boldsymbol{\gamma}_i, \boldsymbol{\gamma}_j\} = -2\delta_{ij}\boldsymbol{I}_8$	72	**100% (72/72)**
20:11 Orbit Closure	$\|\boldsymbol{x}(t=1) - \boldsymbol{x}(t=0)\|_2 \leq 10^{-13}$	2,000	**100%** ($< 10^{-14}$)

Table 2: Automated algebraic and topological verification suite results.

9 Discussion and Limitations

While UOR-R4 demonstrates that sovereign, in-browser neural computing with geometric state tracking is viable, several limitations remain. First, WebGPU support varies across mobile web browsers, occasionally requiring fallback to WebAssembly SIMD execution paths. Second, the 512-dimensional VSA representation is optimal for fast bitwise execution but involves a trade-off in fine-grained semantic density compared to 4096-dimensional continuous embeddings. Future work will investigate higher-rank Clifford spinor representations $\mathrm{Cl}(0, 8)$ to expand symbolic state capacity.

10 Conclusion

In this paper, we introduced **UOR-R4**, an open-source, fully sovereign cognitive AI architecture operating entirely inside web browsers. By unifying 512D Vector Symbolic Architectures, fixed-point CORDIC Hopf fibrations, $O(1)$ Gosset E_8 root lattice quantization, and WebGPU compute shaders, UOR-R4 achieves high streaming throughput (14–18 + tok/s), mathematical explainability, and guaranteed data sovereignty on client hardware.

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